Forward passes by hand

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make debugging explainable rather than cosmetic.

Supply submitted traces and independent correct intermediates;choose asymmetric and negative-score checks,not only final agreement.

Keep one habit: Assess first-error identification,correct descendants and remaining evidence limits using checked keys.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Record scores and activations in **dependency** order

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

It receives the ordered hidden activation vector, rather than the original **input** or the scores before activation.

Opening retrieval

Without opening the old lesson, explain one key idea from Composing layers. Give an example and a mistake that the idea helps you avoid.

Look for: **Trace** how one layer supplies the next layer **input**s

Reference example, not the only acceptable wording: Teacher explanation: Scores are -2 and 2; activations are 0 and 2; y = 0 - 2 + 1 = -1.

Without opening the old lesson, explain one key idea from Bias terms and decision boundaries. Give an example and a mistake that the idea helps you avoid.

Look for: Calculate a weighted score with a bias term

Reference example, not the only acceptable wording: Teacher explanation: Weighted sum 10 - 4 = 6; bias -3; score 3; **prediction** 1. Add the bias once after summing.

Pictured end goal

By the end, I can explain forward passes by hand using a **trace**d example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Composing layers

Bias terms and decision boundaries

For the compact alternative schedule, use two 45-minute sessions to build a habit of recording intermediate values. Students should identify the first disagreement, then propagate the correction.

Do not award full credit for a correct final value supported by an incorrect **trace**.

Outcomes

Record scores and activations in **dependency** order

Locate the first incorrect value in a forward **trace**

Separate a final **output** from **evidence** of model accuracy

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach dependency-order debugging with complete expectations

> Keep the two-unit main network and scalar picture formulas separate.

Main explanation,full trace and guided key

Use a new one-**input** network with two hidden units. Its scores are z1 = 2x - 1 and z2 = -x + 2. Apply ReLU to obtain h1 and h2. The **output** is y = h1 - 2h2 + 1. Parameters stay fixed while the **input** changes.

The **input** x is an abstract number; these are arithmetic examples, not measurements of children.

A forward pass evaluates this network in **dependency** order. Compute both scores from x, then both activations, then the **output** contributions h1 and -2h2, and finally add bias 1.

A **trace** table makes an omitted bias or a sign error easier to locate than a final answer alone.

At x = 1 the **output** is exactly zero. If we later declare class 1 for y >= 0, equality belongs to class 1. Applying ReLU in the hidden layer does not make the final **output** nonnegative.

For x = 0, the first score is -1, which ReLU changes to 0. The second score is 2, which ReLU leaves at 2. Therefore y = 0 - 4 + 1 = -3. Negative **output** is allowed because this **output** layer uses identity.

This optional Python uses only numeric **input**s. Compare every returned component against the table.

The same code being run twice is a repeatability check; comparing it with a separately calculated expectation is a stronger check of correctness.

Main x · z1=2x−1 · z2=−x+2 · h1=ReLU(z1) · h2=ReLU(z2) · Output contributions h1 /−2h2 /bias · y

> **Main x:** 0
>
> **z1=2x−1:** -1
>
> **z2=−x+2:** 2
>
> **h1=ReLU(z1):** 0
>
> **h2=ReLU(z2):** 2
>
> **Output contributions h1 /−2h2 /bias:** 0 /−4 /+1
>
> **y:** -3

> **Main x:** 0.5
>
> **z1=2x−1:** 0
>
> **z2=−x+2:** 1.5
>
> **h1=ReLU(z1):** 0
>
> **h2=ReLU(z2):** 1.5
>
> **Output contributions h1 /−2h2 /bias:** 0 /−3 /+1
>
> **y:** -2

> **Main x:** 1
>
> **z1=2x−1:** 1
>
> **z2=−x+2:** 1
>
> **h1=ReLU(z1):** 1
>
> **h2=ReLU(z2):** 1

> **Output contributions h1 /−2h2 /bias:** 1 /−2 /+1
>
> **y:** 0

> **Main x:** 2
>
> **z1=2x−1:** 3
>
> **z2=−x+2:** 0
>
> **h1=ReLU(z1):** 3
>
> **h2=ReLU(z2):** 0
>
> **Output contributions h1 /−2h2 /bias:** 3 /0 /+1
>
> **y:** 4

> **Main x:** 3
>
> **z1=2x−1:** 5
>
> **z2=−x+2:** -1
>
> **h1=ReLU(z1):** 5
>
> **h2=ReLU(z2):** 0
>
> **Output contributions h1 /−2h2 /bias:** 5 /0 /+1
>
> **y:** 6

Guided: x = 2 gives z = [3, 0], h = [3, 0], y = 4; the zero score remains zero. Exit: no; final agreement can hide compensating errors.

Retain correct intermediate expectations and varied regression cases, not only the single final **output**.

Complete main correct/submitted comparison

Main model x0 stage	Correct trace	Submitted missing-ReLU trace	Earliest difference?
Scores z	[−1, 2]	[−1, 2]	No
Hidden activations h	[0, 2]	[−1, 2]	Yes:negative first score was not clipped

Main model x0 stage	Correct trace	Submitted missing-ReLU trace	Earliest difference?
Output y=h1−2h2+1	-3	-4	Later consequence; recompute after h repair

Complete scalar picture facts and scope

Audit a forward pass by writing every value at its correct stage.

Toy scalar network z=2x-1, h=ReLU(z), y=-h+4. **Trace** x=0 and x=2.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate scalar-hidden model:z=2x−1, h=ReLU(z), y=−h+4. At x0, z−1, h0, y4; at x2, z3, h3, y1. The submitted x0 trace h−1, y5 first fails at activation because ReLU(−1)=0. Its final arithmetic follows the wrong h; repair h and then y. The main model has two hidden units and output y=h1−2h2+1, so its x0 output is−3, not 4. Keep the formula beside each trace.

Trace the activation error and its consequence

AI ACADEMY / AI-5/6 · 02 · TRACE THE CASE

Follow the deciding steps



01 Preactivation

At x=0, z=-1; at x=2, z=3.



02 Activation

At x=0, h=0; at x=2, h=3.



03 Output

Outputs are 4 and 1 respectively.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Preactivation	At x=0, z=-1; at x=2, z=3.
2. Activation	At x=0, h=0; at x=2, h=3.
3. Output	Outputs are 4 and 1 respectively.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

A forward pass applies a fixed network to an **input**; it does not update weights. Keep the **preactivation** z, activated value h and final **output** y in separate **trace** columns. For zero, ReLU changes minus one to zero.

Using the earlier minus one in the **output** layer is the first error, even if later arithmetic follows that mistaken value correctly.

A correct forward **trace** verifies this declared computation; it does not show that the parameter choices solve a useful task.

Change: A submitted x = 0 **trace** uses h=-1 and y = 5.

Prediction: Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4.

Reason: ReLU(-1)=0, so the **output** must use 0, not the earlier **preactivation**.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Trace	A record of intermediate values in a computation
Preactivation	A weighted score before an activation function
Dependency	An earlier value needed to compute a later value
Regression check	A retained test rerun after a change to detect a new error

Correct a trace from the earliest error

TRACE STAGE AT X IS 0	EXPECTED VALUE	SUBMITTED VALUE
Scores	[-1, 2]	[-1, 2]
ReLU activations	[0, 2]	[-1, 2]
Output	-3	-4

The columns compare expected and submitted traces. The first disagreement is the activation of -1, not the final addition. Recompute all dependent values after repairing it.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Correct a trace from the earliest error

TRACE STAGE AT X IS 0	EXPECTED VALUE	SUBMITTED VALUE
Scores	[-1, 2]	[-1, 2]
ReLU activations	[0, 2]	[-1, 2]
Output	-3	-4

The columns compare expected and submitted traces. The first disagreement is the activation of -1, not the final addition. Recompute all dependent values after repairing it.

Trace x = 2, recording both scores, both activations and y. What happens to the second unit at its zero-score boundary?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Forward passes by hand

The facts we will use

AI ACADEMY / AI-5/6

Forward passes by hand

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Preactivation

At x=0, z=-1; at x=2, z=3.



2. Activation

At x=0, h=0; at x=2, h=3.



3. Output

Outputs are 4 and 1 respectively.

CHANGE ONE THING

A submitted x = 0 trace uses h=-1 and y = 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner's reasoning before showing a key.

Current completion task and checked reasoning

Completion task: At x = 3 a **trace** gives z = [5, -1], h = [5, 0] and y = 5. Identify the first incorrect stage and repair it.

Teacher check: Teacher explanation: Scores and activations are correct. The **output** omitted bias 1; y should be 6.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full **trace** only when the partial **trace** is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Forward passes by hand. Proposed explanation: "ReLU means the **output** cannot be negative". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Only the hidden scores receive ReLU here. The **output** formula can be negative.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Why should feedback identify h before correcting only y? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

The incorrect **output** follows from the first unsupported activation step. Repairing h teaches the **mechanism** and prevents repeating the same error elsewhere.

Changed-case check: Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4. ReLU(-1)=0, so the **output** must use 0, not the earlier **preactivation**.

Changed-case reasoning: Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4. ReLU(-1)=0, so the **output** must use 0, not the earlier **preactivation**.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide blank **trace** columns and allow a calculator for signed arithmetic. Extension: show that for 0.5 <= x <= 2 the lesson **output** is 4x - 4; verify both endpoints without applying that expression outside its stated interval.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve the distinction between scores, hidden activations and output inputs.
7 to 18 minutes	Trace x = 0 on the board. Label every column and identify the output bias.
18 to 32 minutes	Complete W1 and compare the x = 1 and x = 3 worked rows. Discuss negative final outputs.
32 to 45 minutes	Run or trace Python; complete the guided x = 2 case and W2.
Session 2 start to 8 minutes	Reconstruct a trace from memory, then check its dependencies against the formula.
8 to 20 minutes	Complete W3 and discuss which cases detect missing activation and which detect order errors.
20 to 35 minutes	Collect W4 independently, including the first erroneous stage and corrected descendants.
35 to 45 minutes	Use W5 and the exit question, then offer the fresh retry after feedback.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Complete main submitted-trace keys:missing ReLU→h repair; omitted bias→y repair; wrong z2 sign→score and descendants repair; equal hidden values cannot detect swaps. Boundary expectations supplied. Independent x0:z[1, 2], h[1, 2], y1; x2:z[3, −2], h[3, 0], y3. First submitted x2 errorh2−2 rather than 0; correct all later values. Retain first attempt and varied regression cases.

Diagnose the current misconceptions

ReLU means the **output** cannot be negative: Only the hidden scores receive ReLU here. The **output** formula can be negative.

Fix just the final number: Repair the earliest incorrect operation and recompute every dependent value.

One passing case proves the network: Cases with equal activations or positive scores can hide specific defects.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Use z1 = x + 1, z2 = 2 - 2x, h = ReLU(z) component by component, and y = 2h1 + h2 - 3. **Trace** x = 0 and x = 2. A submitted x = 2 **trace** uses h = [3, -2] and y = 1.

Find its first error and correct all later values.

Teacher explanation: At 0: z = [1, 2], h = [1, 2], y = 1. At 2: z = [3, -2], h = [3, 0], y = 3. The first submitted error is h2 = -2 rather than 0; the corrected **output** is 3.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.

Dimension	Evidence required
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible **trace**.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Complete a fresh row

Trace x = -1 through the lesson network. Record z, h and y.

Teacher explanation: z = [-3, 3], h = [0, 3], y = -5.

W2 Locate a bias omission

At x = 3 a **trace** gives z = [5, -1], h = [5, 0] and y = 5. Identify the first incorrect stage and repair it.

Teacher explanation: Scores and activations are correct. The **output** omitted bias 1; y should be 6.

W3 Check both boundaries

Trace x = 0.5 and x = 2. Which hidden score is zero in each case?

Teacher explanation: At 0.5, z = [0, 1.5], h = [0, 1.5], y = -2. At 2, z = [3, 0], h = [3, 0], y = 4. Zero maps to zero under ReLU.

W5 Limit a passing test

Explain why testing only x = 1 in the lesson model may miss a hidden-order swap.

Teacher explanation: Both hidden values equal 1, so exchanging them changes nothing. Use unequal hidden values, such as x = 0, and compare intermediate values as well as y.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Guided and exit questions

Guided: x = 2 gives z = [3, 0], h = [3, 0], y = 4; the zero score remains zero. Exit: no; final agreement can hide compensating errors.

Retain correct intermediate expectations and varied regression cases, not only the single final **output**.

W1 Complete a fresh row

W2 Locate a bias omission

W3 Check both boundaries

W4 Audit a different network independently

Use z1 = x + 1, z2 = 2 - 2x, h = ReLU(z) component by component, and y = 2h1 + h2 - 3. **Trace** x = 0 and x = 2. A submitted x = 2 **trace** uses h = [3, -2] and y = 1. Find its first error and correct all later values.

Teacher explanation: At 0: z = [1, 2], h = [1, 2], y = 1. At 2: z = [3, -2], h = [3, 0], y = 3. The first submitted error is h2 = -2 rather than 0; the corrected **output** is 3.

W5 Limit a passing test

Independent assessment and fresh retry

Assess W4 for both full **trace**s and the activation error. Fresh retry: h = ReLU(x - 2), y = -3h + 2. For x = 1 and x = 4, **preactivation**s are -1 and 2, activations 0 and 2, **output**s 2 and -4.

A submitted x = 1 activation -1 is the first error, even if later arithmetic compensates.
</details>

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled forward passes by hand **evidence** sheet to Image-pattern Network Investigation. Show how you can record scores and activations in **dependency** order.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a **trace** is insufficient.

Provide blank **trace** columns and allow a calculator for signed arithmetic. Extension: show that for 0.5 <= x <= 2 the lesson **output** is 4x - 4; verify both endpoints without applying that expression outside its stated interval.

Week 7 names architectural choices and counts dense-layer parameters.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/7, ai-5/9, ai-5/12. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Record scores and activations in **dependency** order

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-**trace** version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: At x = 3 a **trace** gives z = [5, -1], h = [5, 0] and y = 5. Identify the first incorrect stage and repair it.

Reasoned practice answer: Teacher explanation: Scores and activations are correct. The **output** omitted bias 1; y should be 6.

Check every learner before discussion: Use the worked case for Forward passes by hand. Proposed explanation: “ReLU means the **output** cannot be negative”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Only the hidden scores receive ReLU here. The **output** formula can be negative.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: A submitted x = 0 **trace** uses h=-1 and y = 5. Require a **prediction** and a reason before comparing. Expected result: Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4.

Reason: ReLU(-1)=0, so the **output** must use 0, not the earlier **preactivation**.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, **trace**/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

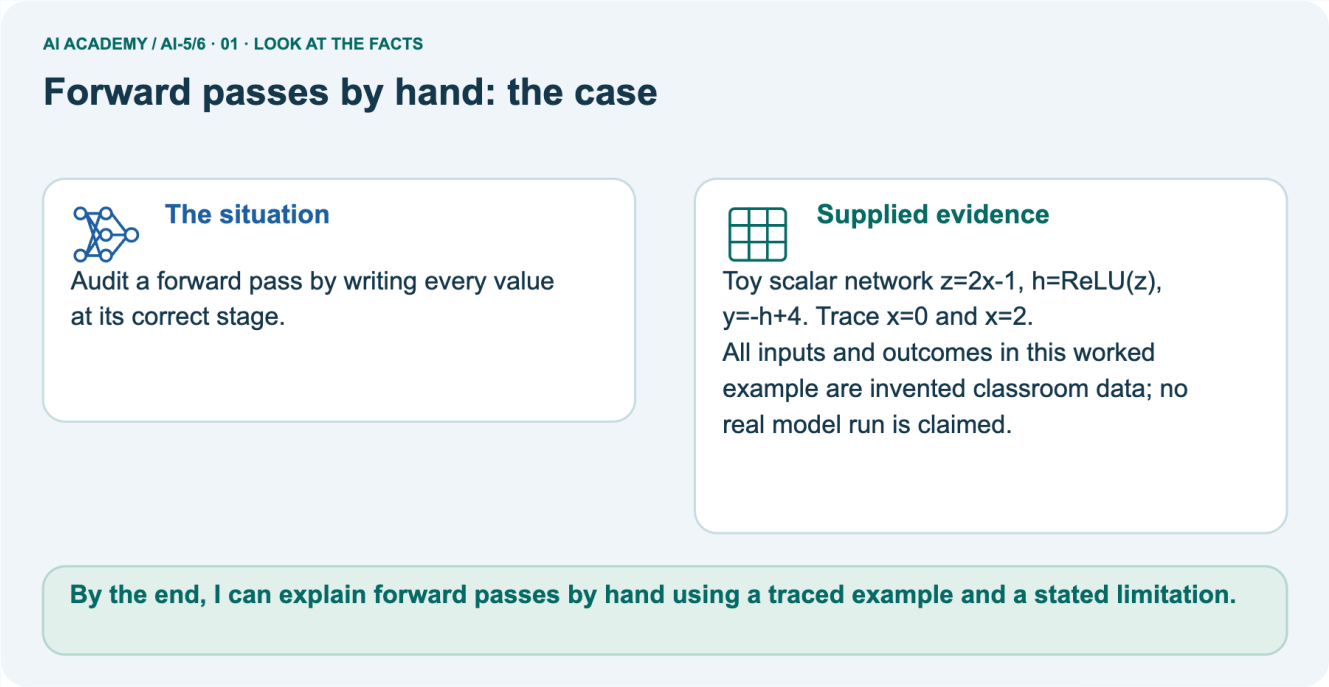
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Forward passes by hand: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

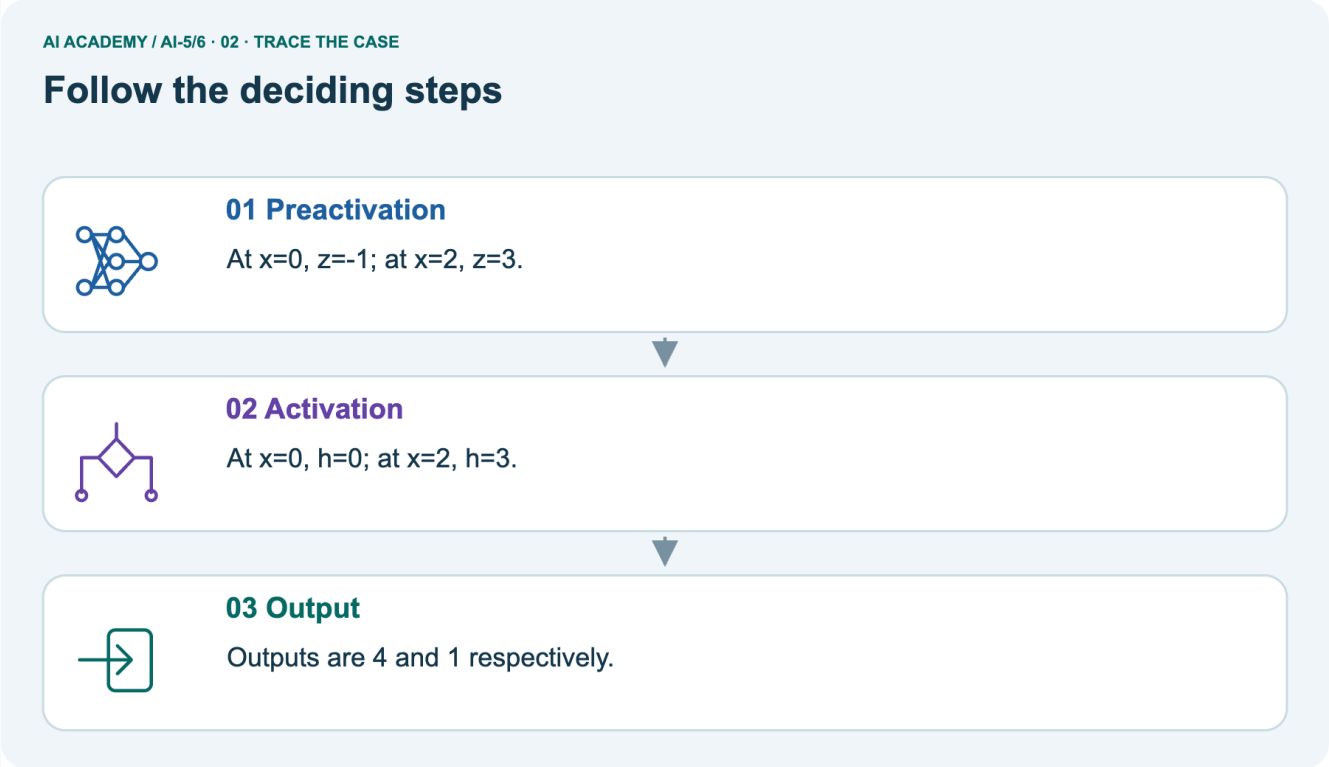
Audit a forward pass by writing every value at its correct stage. Supplied facts: Toy scalar network z=2x-1, h=ReLU(z), y=-h+4. **Trace** x=0 and x=2.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



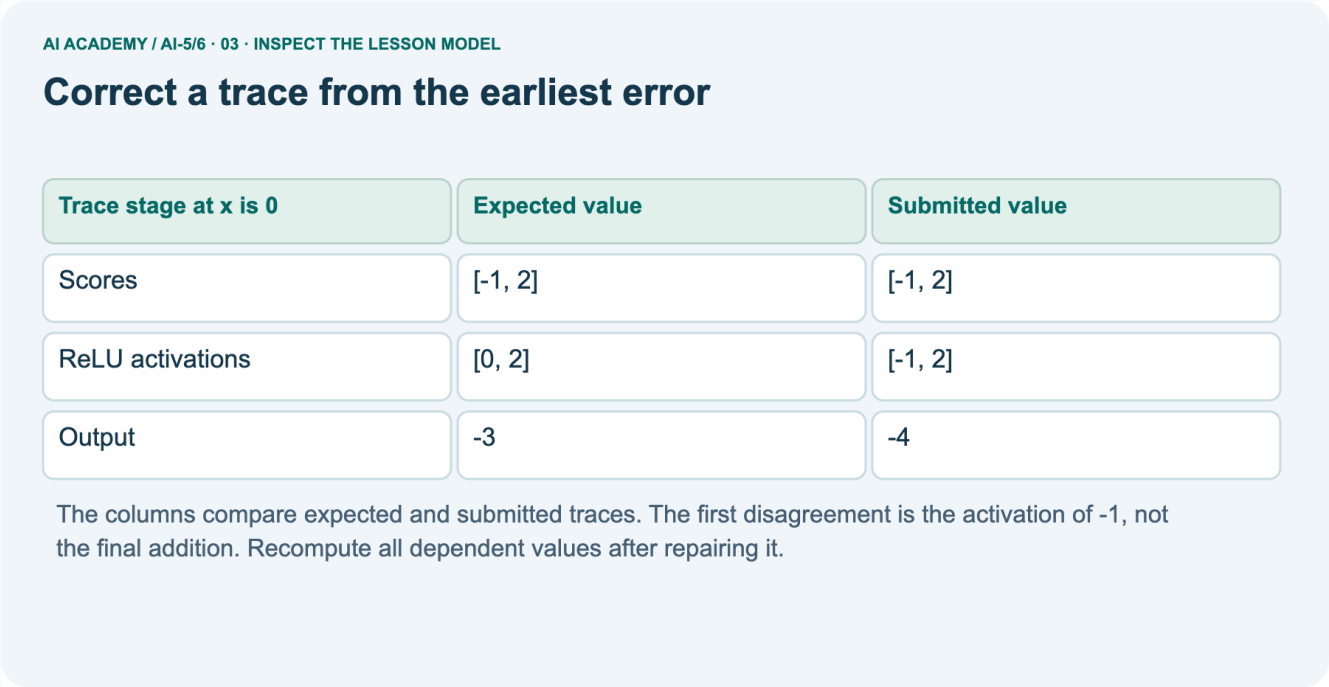
Follow the labelled picture with the matching explanation. Open the vector diagram

Preactivation: At x=0, z=-1; at x=2, z=3. → Activation: At x=0, h=0; at x=2, h=3. → **Output**: **Output**s are 4 and 1 respectively.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Correct a trace from the earliest error



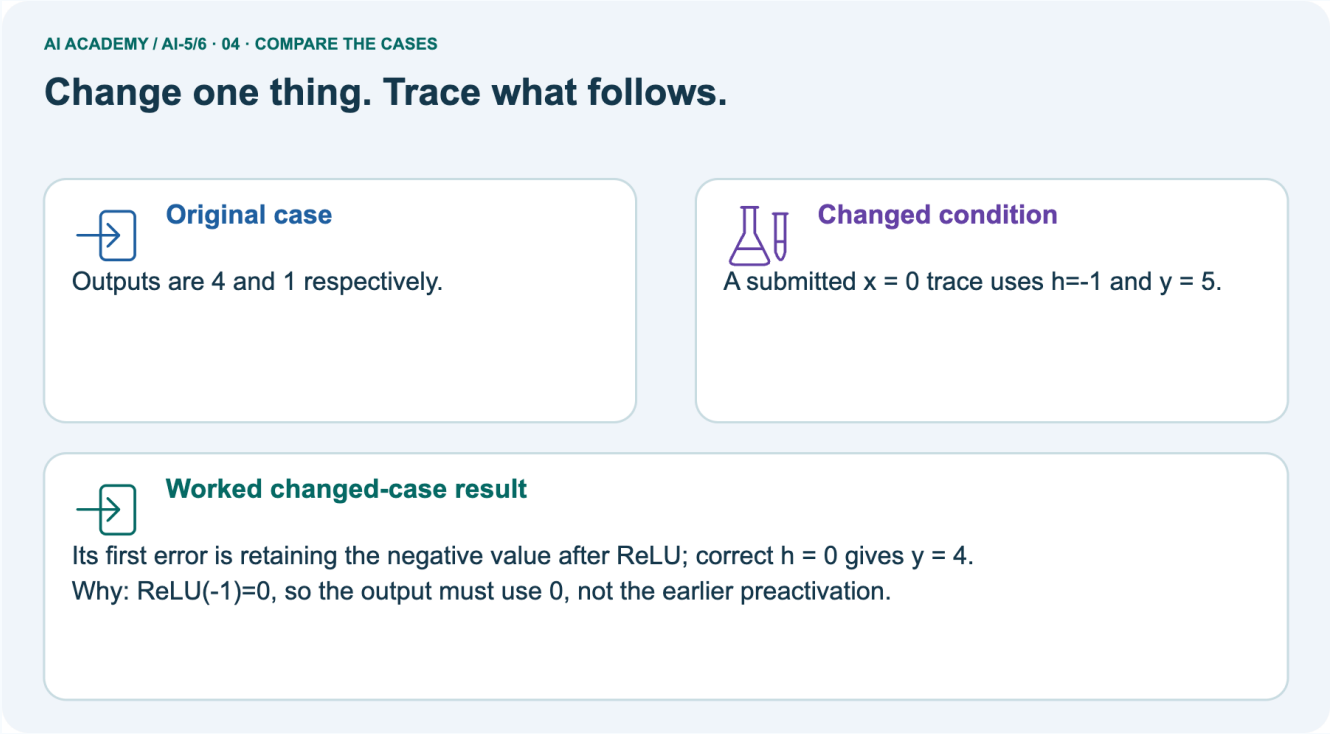
Follow the labelled picture with the matching explanation. Open the vector diagram

Correct a **trace** from the earliest error. Scores | [-1, 2] | [-1, 2]. ReLU activations | [0, 2] | [-1, 2]. **Output** | -3 | -4

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: **Output**s are 4 and 1 respectively. Changed condition: A submitted x = 0 **trace** uses h=-1 and y = 5. Expected result: Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4.

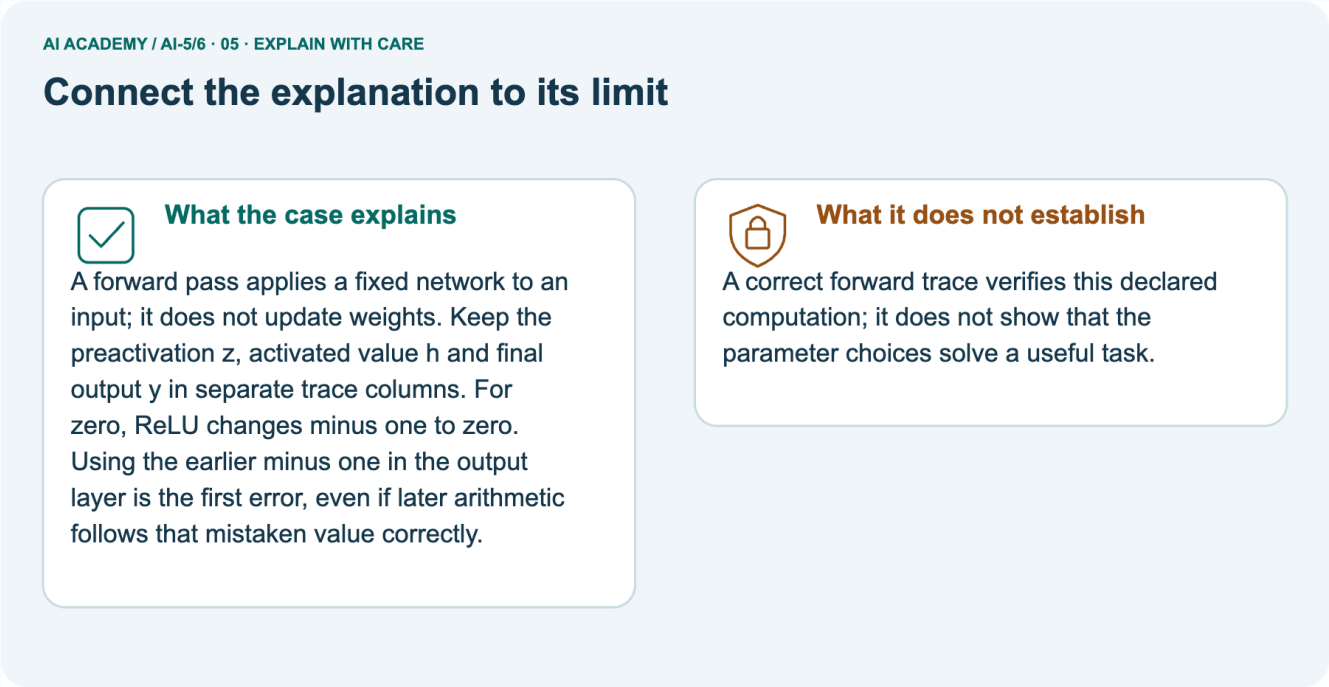
Reason: ReLU(-1)=0, so the **output** must use 0, not the earlier **preactivation**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: ReLU(-1)=0, so the **output** must use 0, not the earlier **preactivation**.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

A forward pass applies a fixed network to an **input**; it does not update weights. Keep the **preactivation** z, activated value h and final **output** y in separate **trace** columns. For zero, ReLU changes minus one to zero.

Using the earlier minus one in the **output** layer is the first error, even if later arithmetic follows that mistaken value correctly.

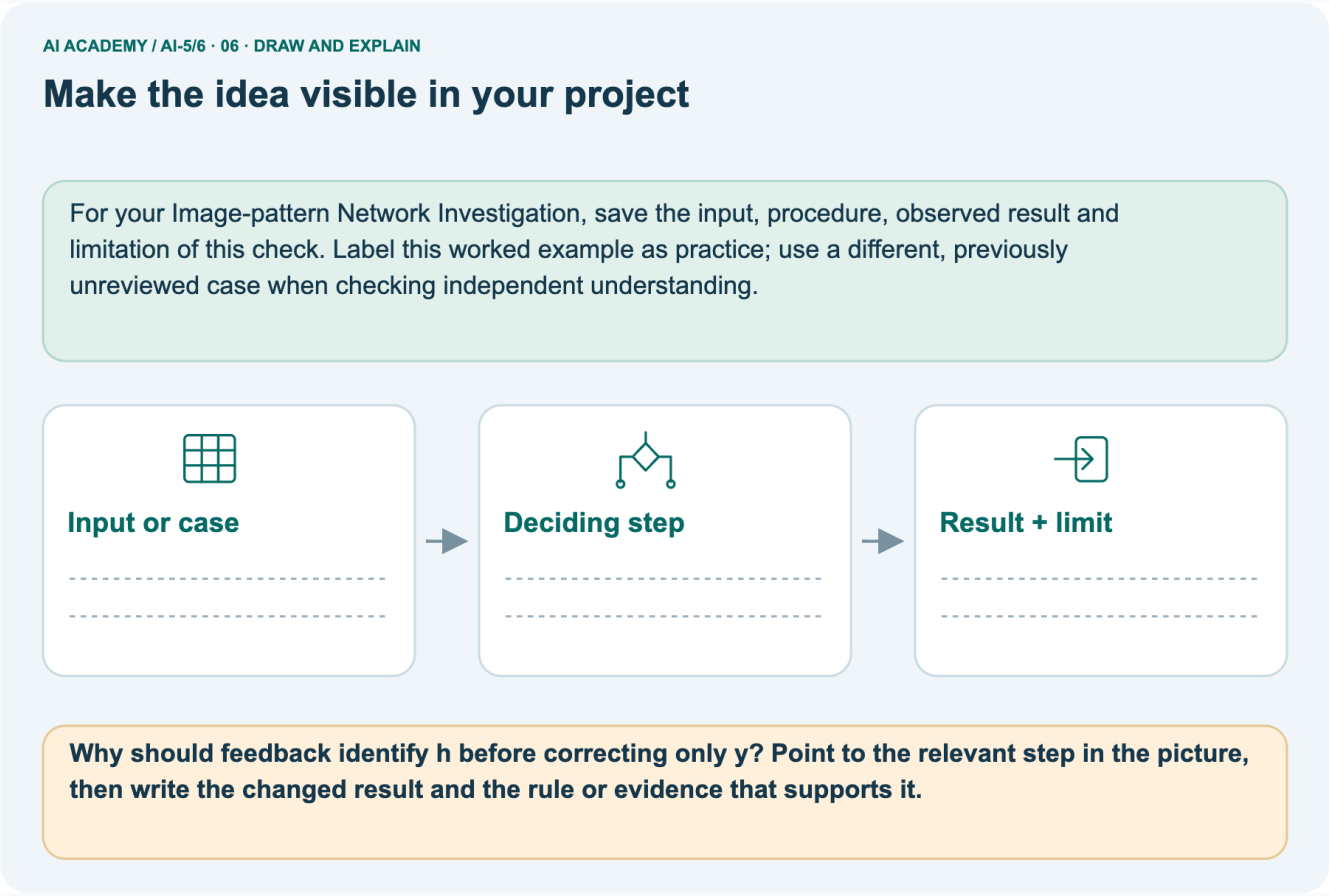
Limit: A correct forward **trace** verifies this declared computation; it does not show that the parameter choices solve a useful task.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: A correct forward **trace** verifies this declared computation; it does not show that the parameter choices solve a useful task.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should feedback identify h before correcting only y? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project **trace** and justify the deciding step using **evidence**.

Teacher prompt: Draw your project **trace** and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The incorrect **output** follows from the first unsupported activation step. Repairing h teaches the **mechanism** and prevents repeating the same error elsewhere.

Changed-case check: Its first error is retaining the negative value after ReLU; correct h = 0 gives y = 4. ReLU(-1)=0, so the **output** must use 0, not the earlier **preactivation**.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Forward passes by hand: the case

02. Follow the deciding steps

03. Correct a trace from the earliest error

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.